

ICAS

PAPER
B



2017 MATHEMATICS

DO NOT OPEN THIS BOOKLET
UNTIL INSTRUCTED.

40 QUESTIONS

TIME ALLOWED: 45 MINUTES

STUDENT'S NAME: _____

Read the instructions on the **ANSWER SHEET** and fill in your **NAME, SCHOOL** and **OTHER INFORMATION**.

Use a pencil. Do **NOT** use a coloured pencil or a pen.

Rub out any mistakes completely.

You **MUST** record your answers on the **ANSWER SHEET**.

Mark only **ONE** answer for each question.

Your score will be the number of correct answers.

Marks are **NOT** deducted for incorrect answers.

There are **40 MULTIPLE-CHOICE QUESTIONS** (1–40).

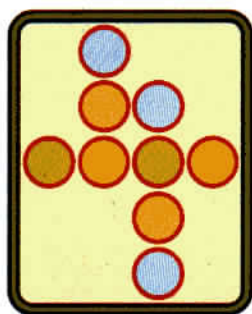
Use the information provided to choose the **BEST** answer from the four possible options.

On your **ANSWER SHEET** fill in the oval that matches your answer.

You may use a ruler and spare paper.

You are **NOT** allowed to use a calculator.

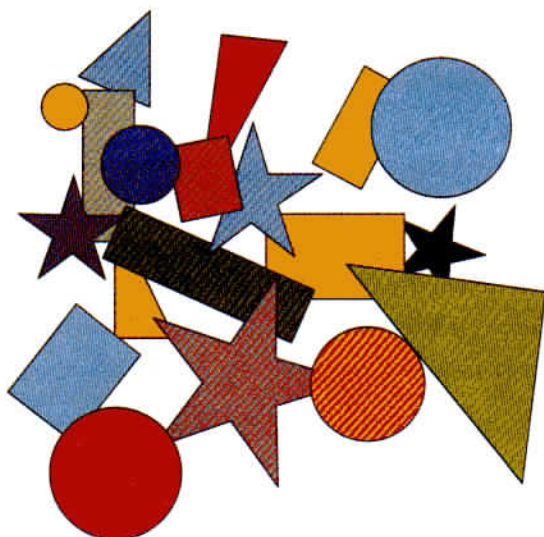
1. Jane put 9 bowls of jelly on a tray.



How many **MORE** bowls of jelly will fit on the tray?

- (A) 9 (B) 11
(C) 16 (D) 20

2. Beth had 19 plastic shapes. There were only triangles, stars, circles and rectangles.



How many triangles did Beth have?

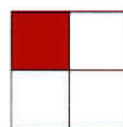
- (A) 1 (B) 2
(C) 3 (D) 4

3. $3 \times 4 = 2 \times ?$

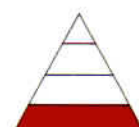
What value must **?** be?

- (A) 4
(B) 5
(C) 6
(D) 12

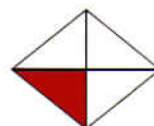
4. Which of these shapes does **NOT** have one-quarter shaded?



(A)



(B)

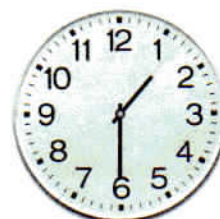


(C)



(D)

5. Arun looked at the clock when he left home for his cricket game.

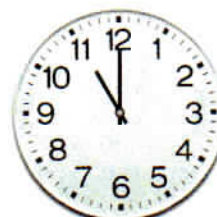


He got back home $3\frac{1}{2}$ hours later.

Which clock shows the time Arun got back home?



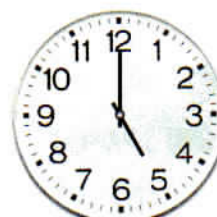
(A)



(B)

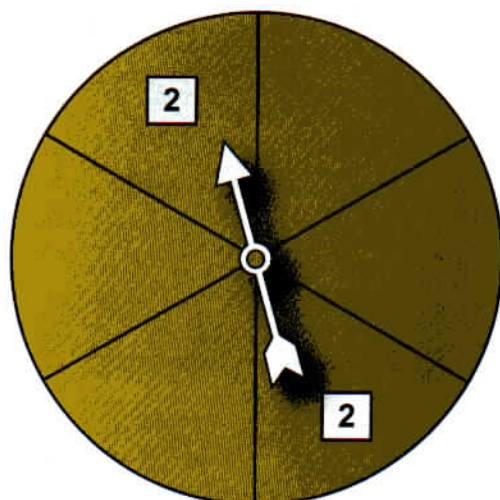


(C)



(D)

6. Sally is making a spinner by placing **2** or **3** in each of the six parts.



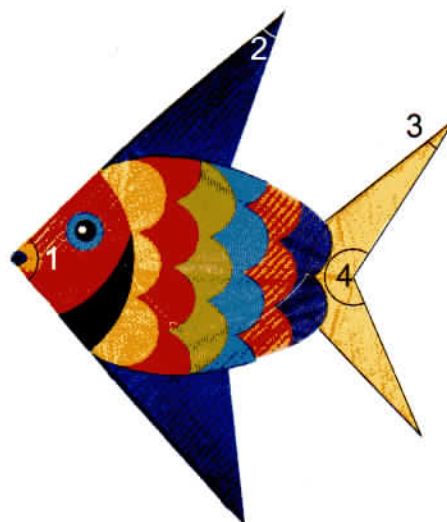
She wants the chances of the spinner landing on a 2 or on a 3 to be the same.

How many more cards with **2** does Sally need to add to her spinner?

- (A) 1
- (B) 2
- (C) 3
- (D) 4

8. Leila had a kite.

She measured some of the angles on the kite.



What is the order of the angles from smallest to largest?

- (A) 3, 2, 1, 4
- (B) 3, 2, 4, 1
- (C) 2, 3, 1, 4
- (D) 2, 3, 4, 1

7. Mal, Sam and Bob are taking turns on a balance board. Mal is heavier than Sam but lighter than Bob.

Which picture is correct?



(A)

(B)



(C)

(D)

9. Cai wrote down a number pattern.

97

93

?

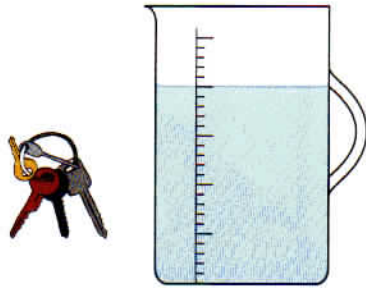
85

81

What number should **?** be?

- (A) 87
- (B) 88
- (C) 89
- (D) 90

10. A teacher dropped his keys in a measuring jug of water.

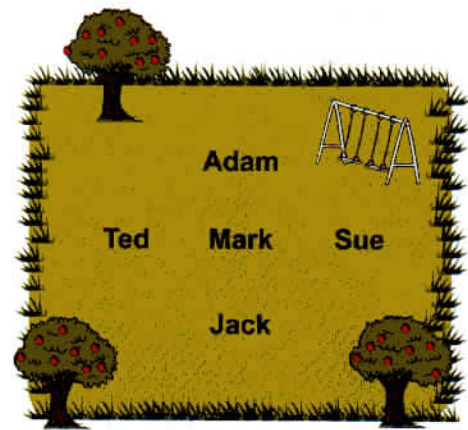


The level of the water rose from 4000 to 4100 on the scale on the jug.

What were the units of these measurements?

- (A) litres
- (B) millilitres
- (C) grams
- (D) kilograms

12. Five students are standing in a park.



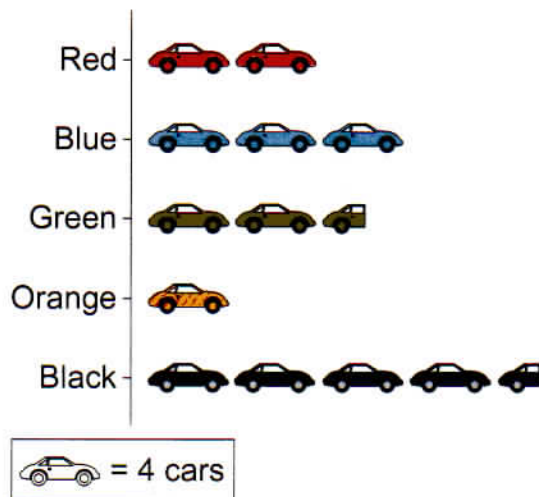
Mark is facing Sue.

Who is standing behind Mark?

- (A) Ted
- (B) Sue
- (C) Jack
- (D) Adam

11. Mr Chipper makes toy cars in different colours.

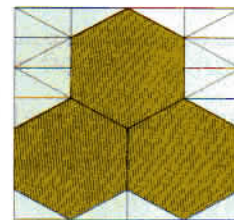
The table shows the number of toy cars Mr Chipper made today.



How many black cars and green cars did Mr Chipper make altogether today?

- (A) 7
- (B) 13
- (C) 28
- (D) 32

13. Lani drew three hexagons on triangular grid paper and shaded them green.



How many triangles did Lani shade?

- (A) 18
- (B) 20
- (C) 30
- (D) 36

14. $32 - 8$ is _____ $16 + 9$.

Which of these makes the statement true?

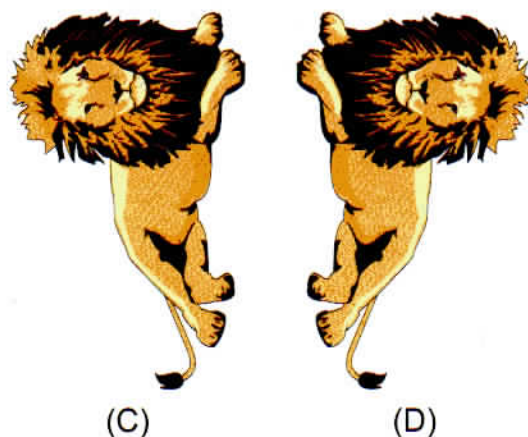
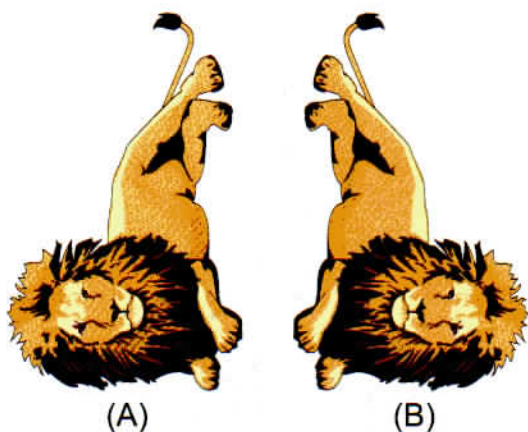
- (A) smaller than
- (B) equal to
- (C) bigger than
- (D) double

15. Mary drew a picture of a lion.



Mary turned it a quarter turn clockwise.

Which of these shows how the picture looked after the turn?



16. Leila is helping her mum to bake cakes.

Each cake needs two cups of flour. For each cup of flour they need three eggs.

How many eggs does Leila need to make two cakes?

- (A) 3
- (B) 5
- (C) 6
- (D) 12

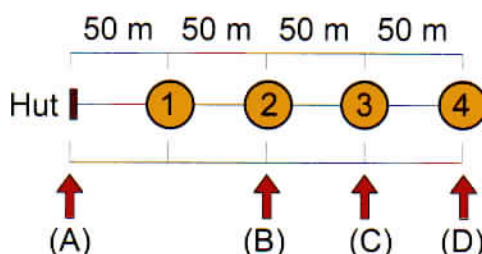
17. A ski lift with cable cars, numbered 1 to 4, has broken down.

Mighty Mitch comes to the rescue.



He flies from the hut and brings the closest cable car back to the hut each time.

Where is Mighty Mitch when he has flown a total of 400 metres?



18. Rachel and Jun were throwing balls at a target. They scored 9 points every time they hit the target.

Rachel hit the target 15 times. She calculated that her score was:

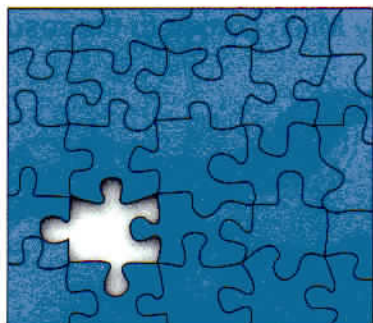
$$15 \times 9 = 135 \text{ points}$$

Jun hit the target 14 times. He used Rachel's score to help him calculate his own total.

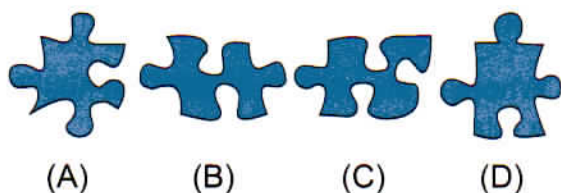
Which calculation gives Jun his score?

- (A) $135 - 15$
- (B) $135 - 14$
- (C) $135 - 9$
- (D) $135 - 1$

19. Jim has lost a piece of his 'Blue Sky' jigsaw puzzle.



Which is the missing piece?



21. Last summer, the Taylors and the Jacksons went to the beach 60 times in total.

The Taylors made 12 more trips to the beach than the Jacksons did.

How many beach trips did the Taylors make?

- (A) 24 (B) 36
(C) 42 (D) 48

22. There are 9 marbles in this bag.



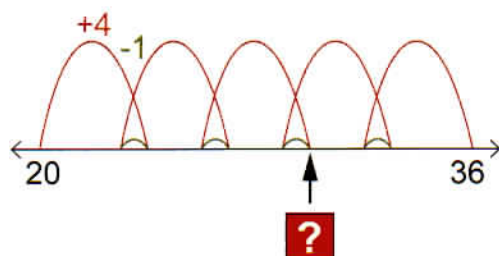
KEY

B = blue
G = green
R = red
Y = yellow

20. Finn made a number pattern with the rule:

add 4, take away 1

He used a number line to show the pattern, starting at 20.



What number must **?** be?

- (A) 32
(B) 31
(C) 30
(D) 29

Tom took out a green marble leaving 8 marbles in the bag.

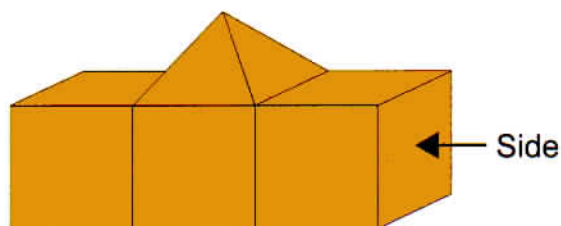
Then it was Julie's turn to take out a marble without looking.

Which two colours does Julie have the same chance of taking out?

- (A) yellow and red
(B) green and red
(C) green and yellow
(D) red and blue

23. Larry built a solid using three cubes and a pyramid.

He drew the view from the side shown.



Which of these should be Larry's drawing?

- (A)
- (B)
- (C)
- (D)

24. Greg weighed 9 identical buttons.

They weighed 27 grams.



How much will 5 buttons weigh?

- (A) 9 grams
(B) 13 grams
(C) 15 grams
(D) 20 grams

25. Faye thinks of a four-digit number.

- The thousands digit is 3 times the tens digit.
- The hundreds digit is even.
- The units digit is an odd number less than 5.

Which of the following could be Faye's number?

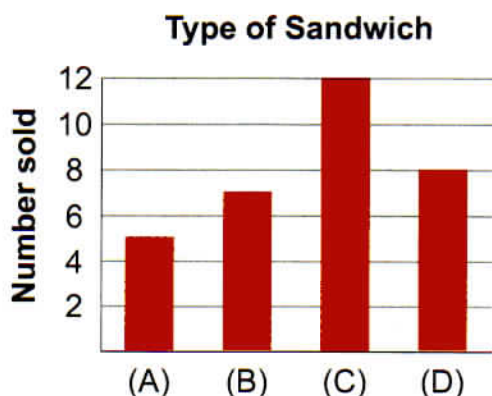
- (A) 1933
(B) 2663
(C) 6425
(D) 9831

26. The canteen sells only fish, cheese, egg or salad sandwiches.

The canteen sold five more fish sandwiches than cheese sandwiches.

Fewer egg sandwiches were sold than salad sandwiches.

Which column represents the number of egg sandwiches sold?



28. Emma is making a pattern of diamonds using small triangles.



Shape 1
2 triangles



Shape 2
8 triangles



Shape 3
18 triangles

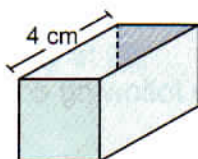


Shape 4

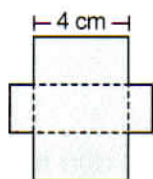
How many small triangles will she need to make the 4th shape in her pattern?

- (A) 32
(B) 28
(C) 25
(D) 22

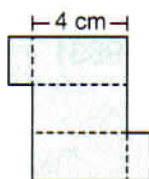
27. This picture shows a box without a lid.



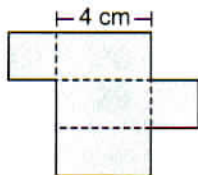
Which of these shows the net for this box?



(A)



(B)

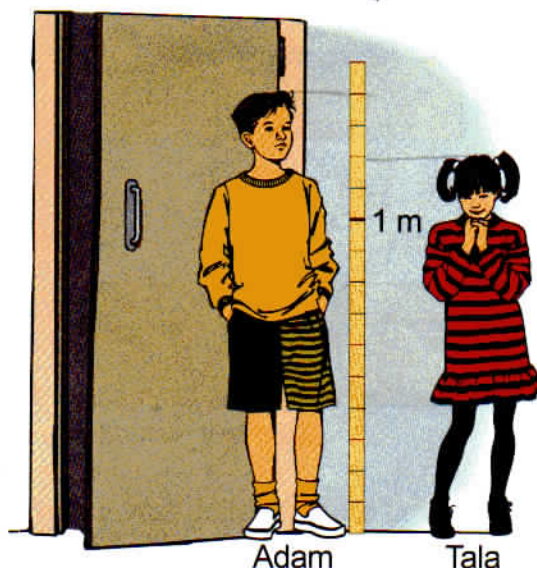


(C)



(D)

29. Using the ruler in the picture, how much taller than Tala is Adam?

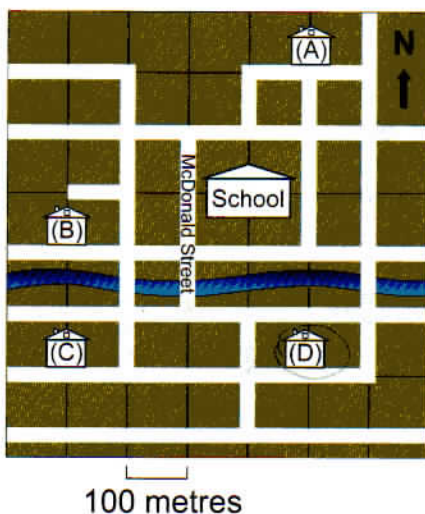


- (A) 2 centimetres
(B) 20 centimetres
(C) 2 metres
(D) 20 metres

30. Freya left this note with the directions to her house.

Leave school, turn left onto McDonald Street and walk to the end. Turn right and walk 100 metres. Turn left and walk 100 metres then turn right and walk 100 metres. My house is the white one.

Which house is Freya's?



31. Sara wrote down three different numbers less than 10. The sum of these numbers was 23.

What was the difference between the largest number and the smallest number?

- (A) 3
- (B) 4
- (C) 5
- (D) 6

32. Ravi takes 4 paces for every 5 of Isha's paces.

Ali takes 3 paces for every 4 of Isha's paces.

Ravi measures the length of the path with 16 paces.

How many paces will Ali take to measure the same path?

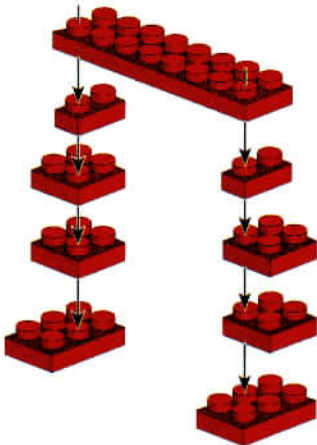
- | | |
|--------|--------|
| (A) 12 | (B) 15 |
| (C) 17 | (D) 20 |

33. Lin calculated $89 - 45 + 76$.

Which of these will give the same answer?

- (A) $89 - 76 + 45$
- (B) $89 - 46 + 75$
- (C) $86 - 48 + 76$
- (D) $86 - 45 + 79$

34. These are the instructions for building an object out of blocks.



What will the object look like when built correctly?

- (A)
- (B)
- (C)
- (D)

35. Jan competes in a running event on the second Friday of each month.

Jan has a running event on Friday 12th April.

What is the date of Jan’s running event in May that same year?

- (A) 3rd May
- (B) 8th May
- (C) 9th May
- (D) 10th May

36. There are 24 students in Mr Sim’s class.

Four students play two sports. All other students play only one sport.

Mr Sim is making a table to show the number of students who play each sport.

Sport	Number
Tennis	
Soccer	8
Volleyball	5
Basketball	?

One-quarter of the students play tennis.

How many students in Mr Sim’s class play basketball?

- (A) 9
- (B) 8
- (C) 6
- (D) 5

37.

$$\begin{array}{r}
 \triangle \star \\
 \star 7 \\
 + \quad 1 \triangle \\
 \hline
 10 \star
 \end{array}$$

- (A) $\triangle = 2$ $\star = 6$
 (B) $\triangle = 3$ $\star = 6$
 (C) $\triangle = 3$ $\star = 5$
 (D) $\triangle = 4$ $\star = 5$

38. Sharon made this solid from seven cubes.



Keeping the cube marked with the white symbol facing her, she rotated the solid clockwise by a quarter of a turn.

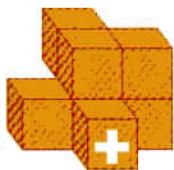
Which diagram shows the result?



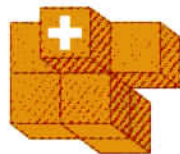
(A)



(B)



(C)



(D)

39. James bought 60 items for his party.



hat
\$1



lantern
\$4



balloon
\$2

One-third of the items were hats.

James bought ten more balloons than lanterns.

How much did James spend altogether?

- (A) \$150
 (B) \$140
 (C) \$130
 (D) \$120

40. Tao, Jin and Lien swam laps in a pool.

Tao started swimming at 3:00 pm.

Jin started swimming at 3:20 pm.

Lien started swimming at 3:25 pm.

They all swam 2 laps per minute and stopped swimming at the same time.

Altogether they swam 240 laps.

At what time did they stop swimming?

- (A) 3:40 pm
 (B) 3:55 pm
 (C) 4:05 pm
 (D) 4:15 pm

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Indonesia	Year 5
Malaysia	Standard 4
Middle East²	Class 4
New Zealand/Pacific³	Year 5
Singapore	Primary 3
Southern Africa⁴	Grade 4

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